Contents

1 字串

1.1 字典樹 trie

```cpp
struct trie{
    trie *nxt[26]={};
    int cnt=0;
    int sz=0;
};
trie *root=new trie();
void insert(const string &s){
    trie *now=root;
    for(auto i:s){
        if(now->nxt[i-'a']==NULL){
            now->nxt[i-'a']=new trie();
        }
        now=now->nxt[i-'a'];
}}
int qry(string &s){
    trie *now=root;
    for(auto i:s){
        if(now->nxt[i-'a']==NULL) return;
        now=now->nxt[i-'a'];
    }
    return now->cnt;
}
```

1.2 KMP

```cpp
vector<int> kmp(string s,string t){
    if(t.size()>s.size()) return {};
    vector<int> pi(t.size(),0);
    for(int i=1,j=0;i<t.size();i++){
        while(j>0 && t[j]!=t[i]){
            j=pi[j-1];
        }
        if(t[j]==t[i]) j++;
        pi[i]=j;
    }
    vector<int> ans;
    for(int i=0,j=0;i<s.size();i++){
        while(j>0 && t[j]!=s[i]){
            j=pi[j-1];
        }
        if(t[j]==s[i]) j++;

        if(j==t.size()){
            ans.push_back(i-j+1);
            j=pi[j-1];
    }}
    return ans;
}
```

2 最短路徑

2.1 basic

```cpp
// c++ code
#include <bits/stdc++.h>
using namespace std;

int main() {
    // test comment
    cout << "test string\n";
}
```

3 圖論

3.1 凸包問題

```cpp
struct node{
    int x,y;
    bool operator<(const node &other) const{
        return
        (x<other.x) || (x==other.x && y<other.y);
    };
};
struct ConvexHull{
    vector<node> vec;
    int n;
    void init(vector<node> &v){
        vec=v;
        n=v.size();
    }
    int cross(node a,node o,node b){
        return
        (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
    }
    vector<int> point;
    vector<int> area;
    void build(){
        point=vector<int>(n+1);
        area=vector<int>(n+1);
        vector<node> up,down;
        int upsum=0;
        int downsum=0;
        for(int i=0;i<n;i++){
            while(up.size()>=2 &&
            cross(up[up.size()-2],
                  up[up.size()-1],
                  vec[i])<=0){
                upsum-=cross(up[up.size()-2],
                             {0,0},up[up.size()-1]);
                up.pop_back();

            }
            up.push_back(vec[i]);
            if(up.size()>1)
                upsum+=cross(up[up.size()-2],
                             {0,0},
                             up[up.size()-1]);

            while(down.size()>=2 &&
            cross(down[down.size()-2],
                  down[down.size()-1],
                  vec[i])>=0){
                downsum-=cross(down[down.size()-2],
                               {0,0},down[down.size()-1]);
                down.pop_back();
            }
            down.push_back(vec[i]);
            if(down.size()>1)
                downsum+=cross(down[down.size()-2],
```

```
                                        {0,0},down[down.size()-1]);

            area[i+1]=abs(upsum-downsum);

            if(i+1==1) point[i+1]=1;
            else point[i+1]=up.size()+down.size()-2;
    }}
    double getarea(int p){
        return area[p]/2.0;
    }
    int getnum(int p){
        return point[p];
    }
};
```

4 數學

4.1 公式

- Lucas's Theorem :
  For $n, m \in \mathbb{Z}^*$ and prime P, $C(m,n) \mod P = \Pi(C(m_i, n_i))$ where m_i is the i-th digit of m in base P.

- Stirling approximation :
  $n! \approx \sqrt{2\pi n}(\frac{n}{e})^n e^{\frac{1}{12n}}$

- Stirling Numbers(permutation $|P| = n$ with k cycles):
  $S(n,k) =$ coefficient of x^k in $\Pi_{i=0}^{n-1}(x+i)$

- Stirling Numbers(Partition n elements into k non-empty set):
  $S(n,k) = \frac{1}{k!}\sum\limits_{j=0}^{k}(-1)^{k-j}\binom{k}{j}j^n$

- Pick's Theorem : $A = i + b/2 - 1$
  A: Area、i: grid number in the inner、b: grid number on the side

- Catalan number : $C_n = \binom{2n}{n}/(n+1)$
  $C_n^{n+m} - C_{n+1}^{n+m} = (m+n)!\frac{n-m+1}{n+1} \quad for \quad n \ge m$
  $C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$
  $C_0 = 1 \quad and \quad C_{n+1} = 2(\frac{2n+1}{n+2})C_n$
  $C_0 = 1 \quad and \quad C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i} \quad for \quad n \ge 0$

- Euler Characteristic:
  planar graph: $V - E + F - C = 1$
  convex polyhedron: $V - E + F = 2$
  V, E, F, C: number of vertices, edges, faces(regions), and components

- Kirchhoff's theorem :
  $A_{ii} = deg(i), A_{ij} = (i,j) \in E \ ? -1 : 0$, Deleting any one row, one column, and cal the det(A)

- Polya' theorem (c is number of color，m is the number of cycle size):
  $(\sum_{i=1}^{m} c^{gcd(i,m)})/m$

- Burnside lemma:
  $|X/G| = \frac{1}{|G|}\sum\limits_{g\in G}|X^g|$

- 錯排公式: (n 個人中，每個人皆不再原來位置的組合數):
  $dp[0] = 1; dp[1] = 0;$
  $dp[i] = (i-1) * (dp[i-1] + dp[i-2]);$

- Bell 數 (有 n 個人, 把他們拆組的方法總數) :
  $B_0 = 1$
  $B_n = \sum_{k=0}^{n} s(n,k) \quad (second - stirling)$
  $B_{n+1} = \sum_{k=0}^{n}\binom{n}{k}B_k$

- Wilson's theorem :
  $(p-1)! \equiv -1(mod\ p)$

- Fermat's little theorem :
  $a^p \equiv a(mod\ p)$

- Euler's totient function:
  $A^{B^C} \ mod\ p = pow(A, pow(B, C, p-1))mod\ p$

- 歐拉函數降冪公式:
  $A^B \mod C = A^{B \mod \phi(c) + \phi(c)} \mod C$

- 6 的倍數:
  $(a-1)^3 + (a+1)^3 + (-a)^3 + (-a)^3 = 6a$

- Standard young tableau (標準楊表):
  $\lambda = (\lambda_1 \ge \cdots \ge \lambda_k), \sum \lambda_i = n$ denoted by $\lambda \vdash n$
  $\lambda \vdash n$ 意思為 λ 整數拆分 n eg. $n = 10, \lambda = (6,4)$ 此拆分可表示一種楊表形狀。
  楊表: 第 1 列 λ_1 行 $\cdots$ 第 k 列 λ_k 行的方格圖。
  標準楊表: 每列從左到右遞增，每行從上到下遞增。
  Let T 為某一 Permutation 跑 RSK 後的標準楊表，則此 Permutation 的 LDS、LIS 長度分別為 T 的列、行數。

- RSK Correspondence:
  A permutation is bijective to (P,Q) 一對標準楊表
  P : Permutation 跑 RSK 算法的結果，可為半標準楊表。
  Q : 可用來還原 Permutation (像排列矩陣)。

- Hook length formula (形狀為 λ 的標準楊表個數):
  $f^\lambda = \frac{n!}{\prod h_\lambda(i,j)}$
  $h_\lambda(i,j) =$ number of pair (x,y) where $(x = i \vee y = j) \wedge (x,y) \ge (i,j)$ 且 (x,y) 落在形狀為 λ 的表上。
  Recursion:
  (i) $f^{(0,\cdots,0)} = 1$
  (ii) $f^{(\lambda_1,\cdots,\lambda_m)} = \sum_{k=1}^{m} f^{(\lambda_1,\cdots,\lambda_{k-1},\lambda_k-1,\lambda_{k+1},\cdots,\lambda_m)}$

- 一個環上相鄰顏色不同的染色數量:
  $f(n,k) = (k-1)^n + (-1)^n(k-1)$
  n 是點的數量，k 是顏色的數量